



IIC3670 Procesamiento de Lenguaje Natural

<https://github.com/marcelomendoza/IIC3670>

- PROMPTING -

Prompting

Dado el prompt, el LLM generara texto condicionado al prompt. Usualmente el prompt es de autocompletación.

Hotel Review Completions

Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relaxed. In short our stay was

... not a pleasant one. The staff at the front desk were not welcoming or friendly, and seemed disinterested in providing good customer service.

... uncomfortable and not worth the price we paid. We will not be returning to this hotel.

Prompting

Los prompts pueden especificar instrucciones. Los LLMs han sido alineados para seguir varias instrucciones. Aquí algunos ejemplos:

Basic Prompt Templates

Summarization	<code>{input} ; tldr;</code>
Translation	<code>{input} ; translate to French:</code>
Sentiment	<code>{input}; Overall, it was</code>
Fine-Grained-Sentiment	<code>{input}; What aspects were important in this review?</code>

Prompting

Dar estructura al prompt ayuda al LLM a reconocer la instrucción, ejemplos, alternativas e incluso roles.

A prompt consisting of a review plus an incomplete statement

Human: Do you think that “input” has negative or positive sentiment?

Choices:

(P) Positive

(N) Negative

Assistant: I believe the best answer is: (

In-context learning

Podemos mostrar algunos ejemplos de la tarea que queremos resolver en el prompt. A esto se le denomina *in-context learning*:

Few shot learning (k=3) →

Definition: This task is about writing a correct answer for the reading comprehension task. Based on the information provided in a given passage, you should identify the shortest continuous text span from the passage that serves as an answer to the given question. Avoid answers that are incorrect or provides incomplete justification for the question.

Passage: Beyoncé Giselle Knowles-Carter (born September 4, 1981) is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, *Dangerously in Love* (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

Examples:

Q: In what city and state did Beyoncé grow up?

A: Houston, Texas

Q: What areas did Beyoncé compete in when she was growing up?

A: singing and dancing

Q: When did Beyoncé release *Dangerously in Love*?

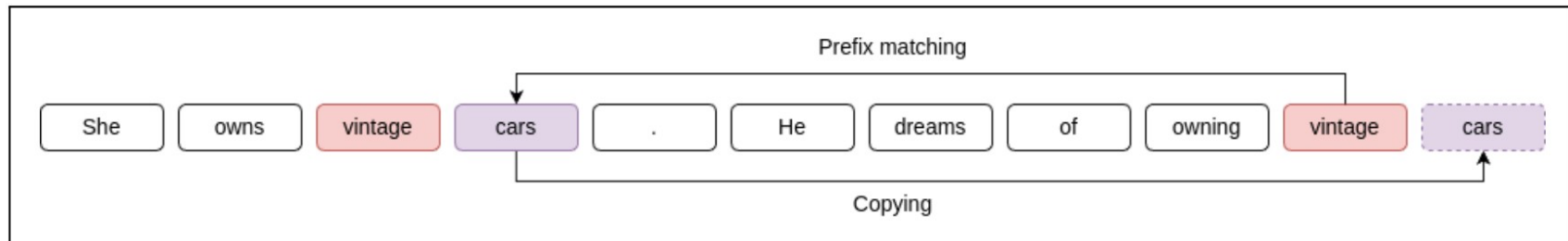
A: 2003

Q: When did Beyoncé start becoming popular?

A:

In-context learning

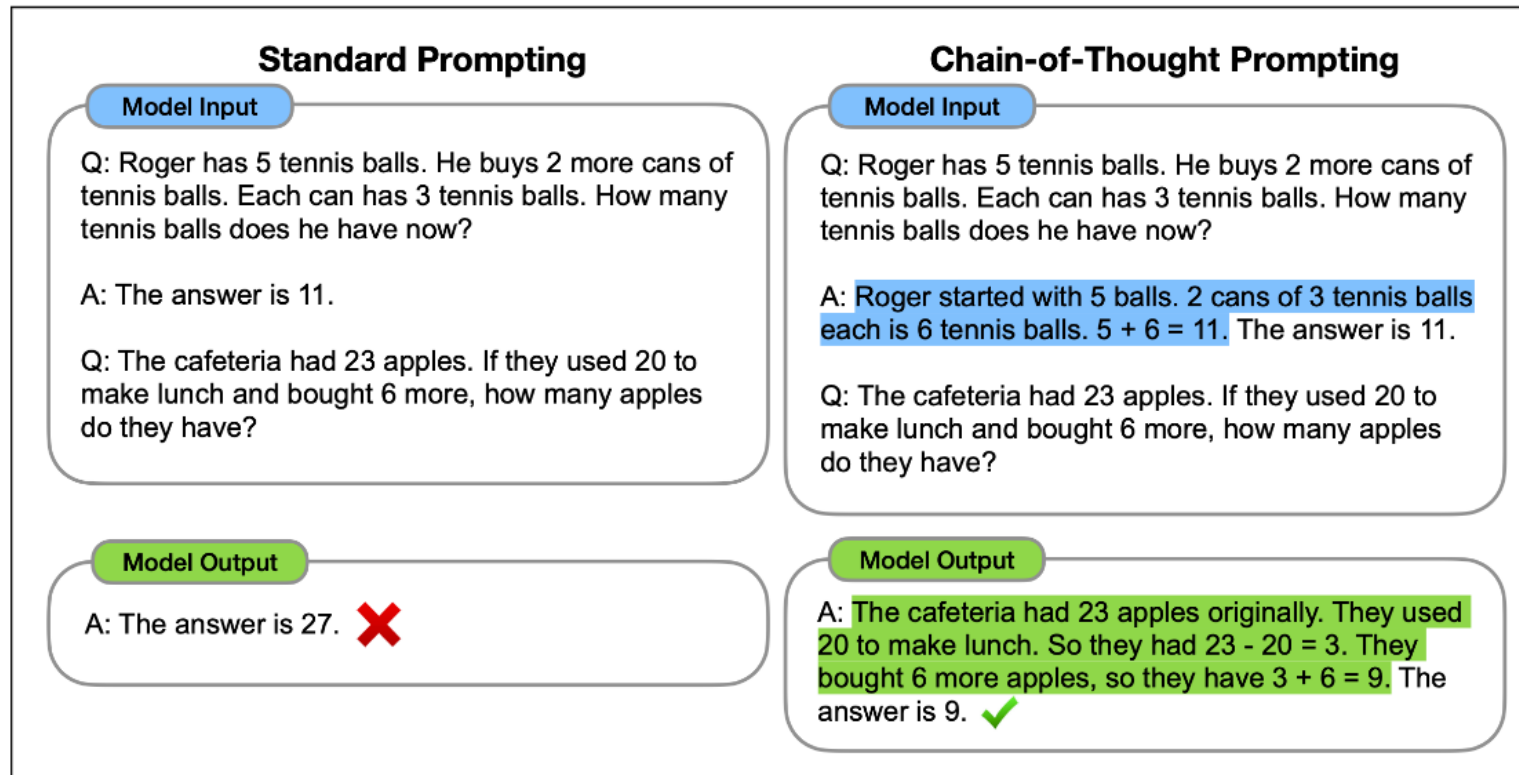
Los heads inductivos adquieren la capacidad de predecir secuencias repetidas.



Usa un mecanismo de copia para inferir la salida sobre la base de los datos de preentrenamiento.

Chain-of-Thought (CoT)

Incluir un ejemplo razonado (pasos para encontrar la solución) es más informativo que sólo incluir un ejemplo con la respuesta esperada.



El LLM, al imitar el procedimiento, tiene más herramientas para encontrar la solución.



Wei, J., X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. NeurIPS, volume 35

ReAct

Combinar razonamiento + acciones (por ejemplo, búsqueda en fuentes externas), permite abordar problemas más difíciles.

(1) Hotspot QA

Question: How many rooms are in the hotel that is home to the Cirque du Soleil show Mystere?

HotpotQA label: 2,664

Outdated X

(1a) Standard

Answer: 3,000

X

(1b) CoT (Reason Only)

Thought: Let's think step by step. The hotel that is home to the Cirque du Soleil show Mystere is Treasure Island. Treasure Island has 2,885 rooms, so the answer is 2,885.

Answer: 2,885

X

(1c) Act-Only

Act 1: Search[Cirque du Soleil]

Obs 1: Cirque du Soleil is a Canadian entertainment company...

Act 2: Lookup[Mystere]

Obs 2: No more results

Act 3: Search[Mystere (Cirque du Soleil show)]

Obs 3: Could not find...

Act 4: Search[Mystere (Cirque du Soleil)]

Obs 4: Mystère (my-steer) is one of six...

Act 5: Lookup[Treasure Island Hotel and Casino]

.....(end without answer)

X

(1d) ReAct (Reason + Act)

Thought 1: I need to search Cirque du Soleil show Mystere, find the hotel it is in, then find the number of rooms in the hotel.

Act 1: Search[Cirque du Soleil show Mystere]

Obs 1: Could not find Cirque du Soleil show Mystere. Similar: ['Cirque du Soleil' ...]

Thought 2: To find the hotel, I can search Mystere (Cirque du Soleil).

Act 2: Search[Mystere (Cirque du Soleil)]

Obs 2: Mystère (my-steer) is ... Held in a custom theatre at the Treasure Island Hotel and Casino ...

Thought 3: Mystere is held in a custom theatre at the Treasure Island Hotel and Casino. So I need to search Treasure Island Hotel and Casino next and find the number of rooms in the hotel.

Act 3: Search[Treasure Island Hotel and Casino]

Obs 3: Treasure Island Hotel and Casino ... with 2,884 rooms and 220 suites ...

Thought 4: Treasure Island Hotel and Casino has 2,884 rooms and 220 suites. So the answer is 3,104.

Act 4: Finish[3,104]

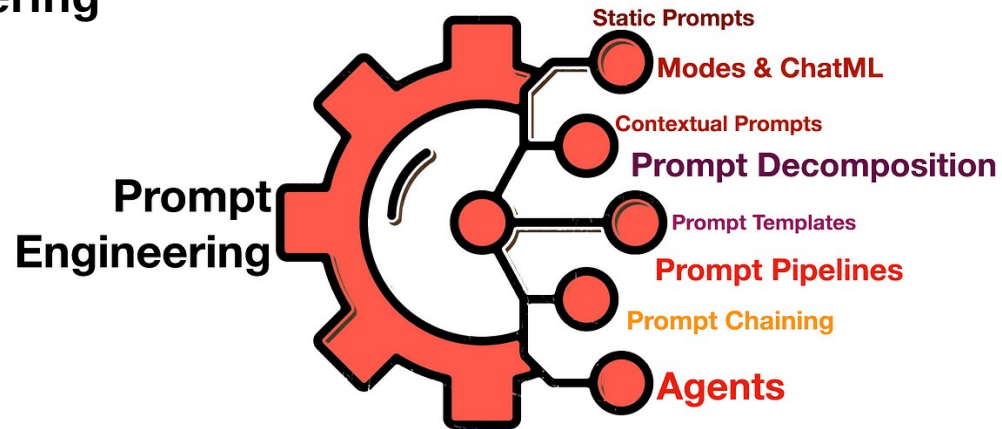
Up-to-date ✓

Si el LLM no tiene información, la va a buscar.

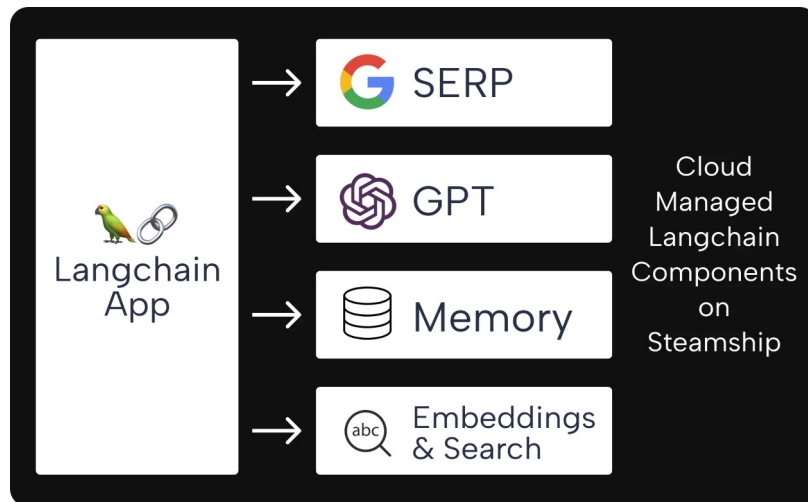


Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023.

Prompt Engineering



LangChain

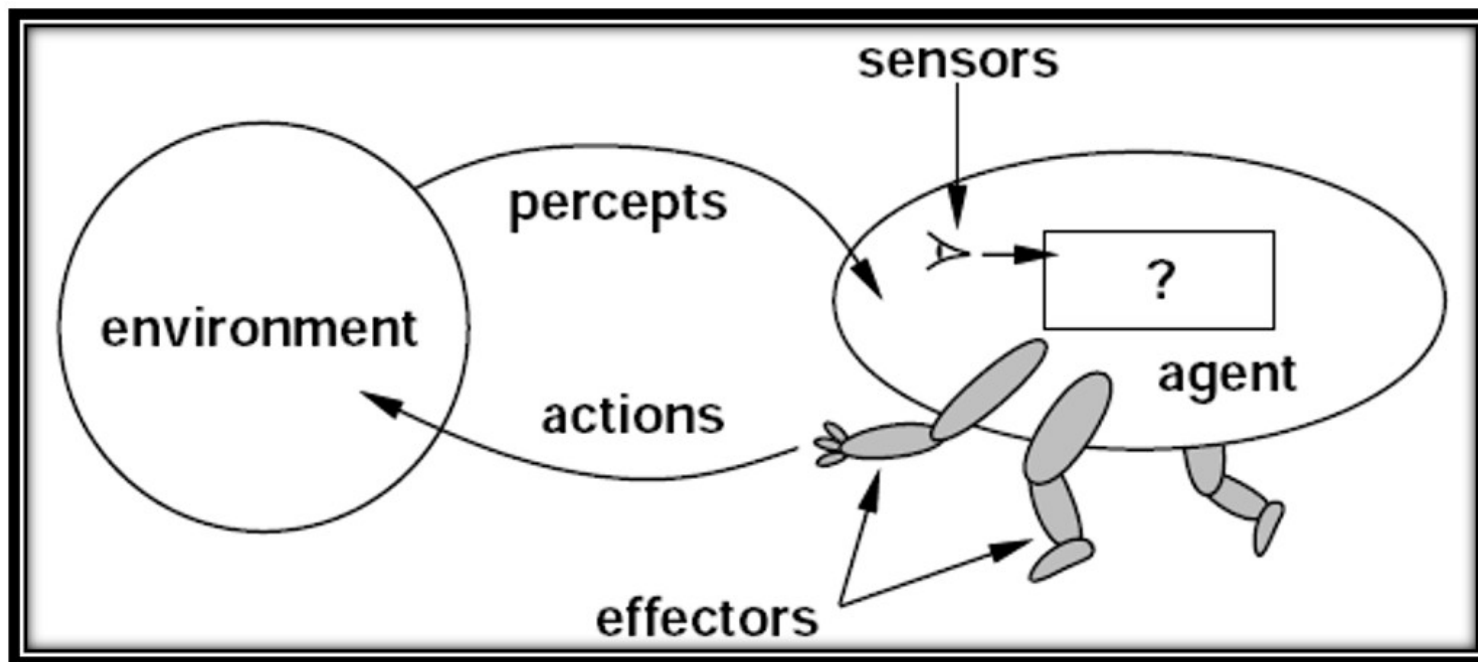


DSPy



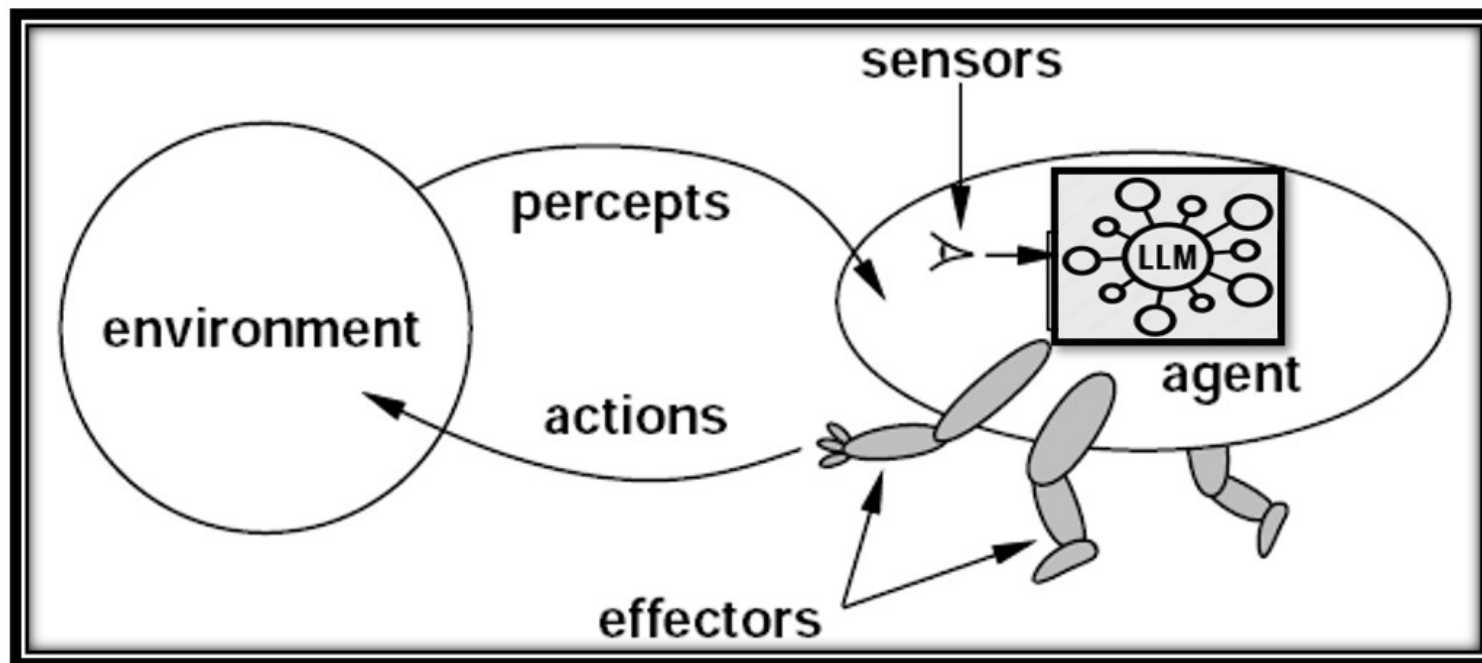
Agentes

Un agente es una entidad (software) que percibe el entorno, procesa información y toma acciones para lograr objetivos y metas específicas.



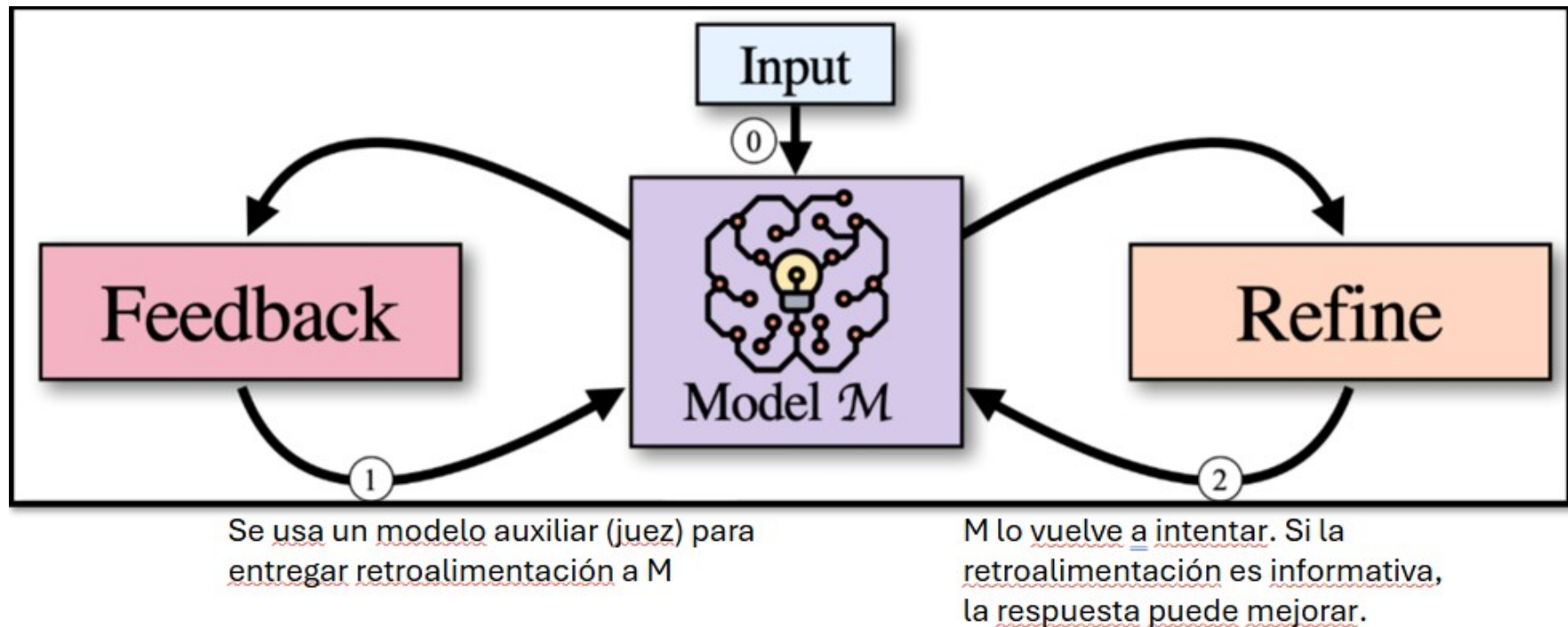
Agentes

Un agente incluye una IA (LLM), percibe el entorno, procesa información y toma acciones para lograr objetivos y metas específicas.



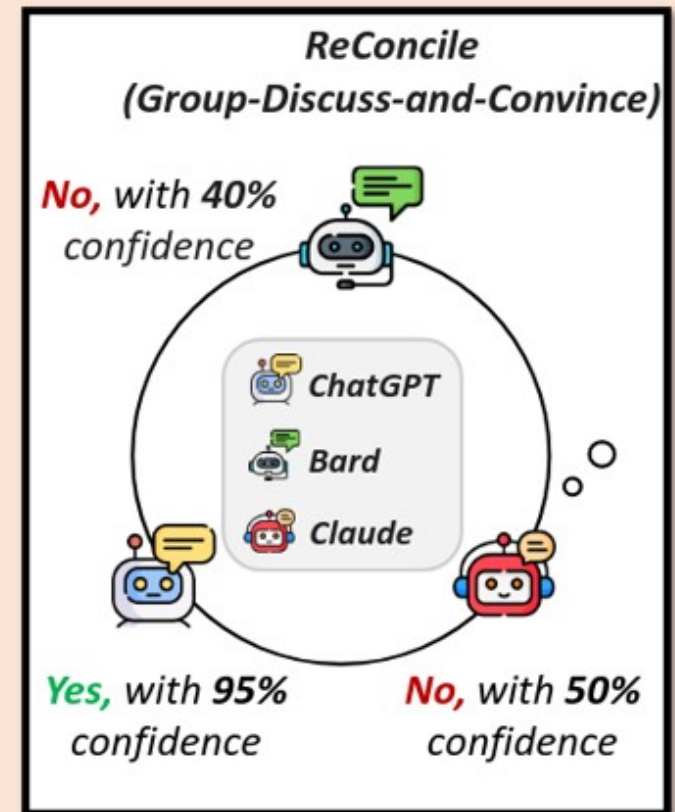
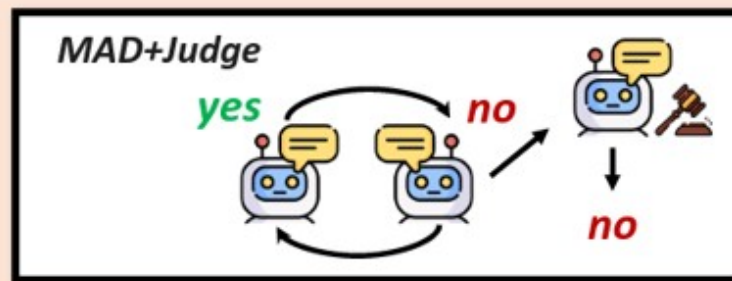
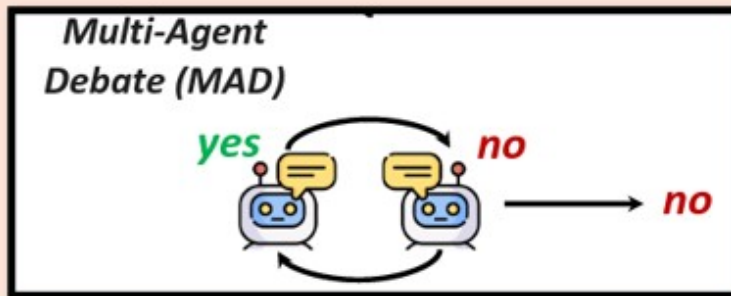
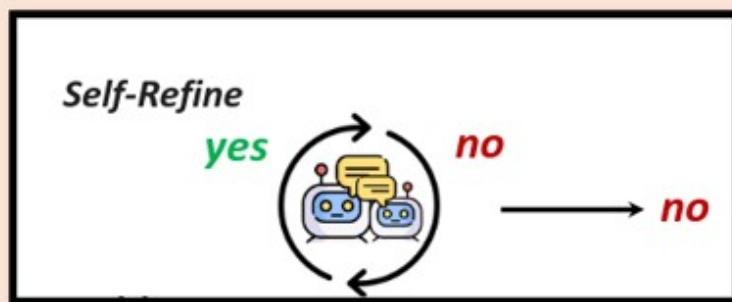
Self-Refine

Un modelo auxiliar (juez) califica la respuesta y entrega feedback. En base al feedback, el modelo mejora su prompt (prompt para hacer prompt).



Estrategias de interacción en agentes

Compartir resultados o incluso su justificación puede ser útil en una estrategia de colaboración.



Estrategias de interacción en agentes

Podemos dividir para conquistar. Dividimos el problema, generamos salida estructurada, el juez toma la salida, aplica la rúbrica, evalúa y consolida.

